SOC Log Analyzer – Project Report

1. Introduction

The **SOC Log Analyzer** is an AI-powered cybersecurity application designed to analyze system and network logs to detect potential security threats. The system leverages **Large Language Models (LLMs)** along with a structured database system to process log data, identify anomalies, and generate meaningful summaries for Security Operations Center (SOC) analysts.

Traditional log analysis systems rely on static rules and manual inspection, which are time-consuming and often ineffective for detecting unknown threats. This project enhances the process by integrating AI and a relational database (**PostgreSQL**) to provide **scalable, intelligent, and efficient threat detection**.

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2. Objective

The primary objectives of this project are:

- Automate analysis of large-scale log data
- Detect suspicious and malicious activities
- Store and manage logs efficiently using PostgreSQL
- Provide clear summaries of critical events
- Assist SOC analysts in faster decision-making

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3. Input Data

The system processes log files from sources such as:

- Zscaler Web Proxy Logs
- Server logs
- Application logs

Each log entry typically includes:

- Timestamp
- IP Address
- URL / Domain
- Action (ALLOW / BLOCK)

- Status Code

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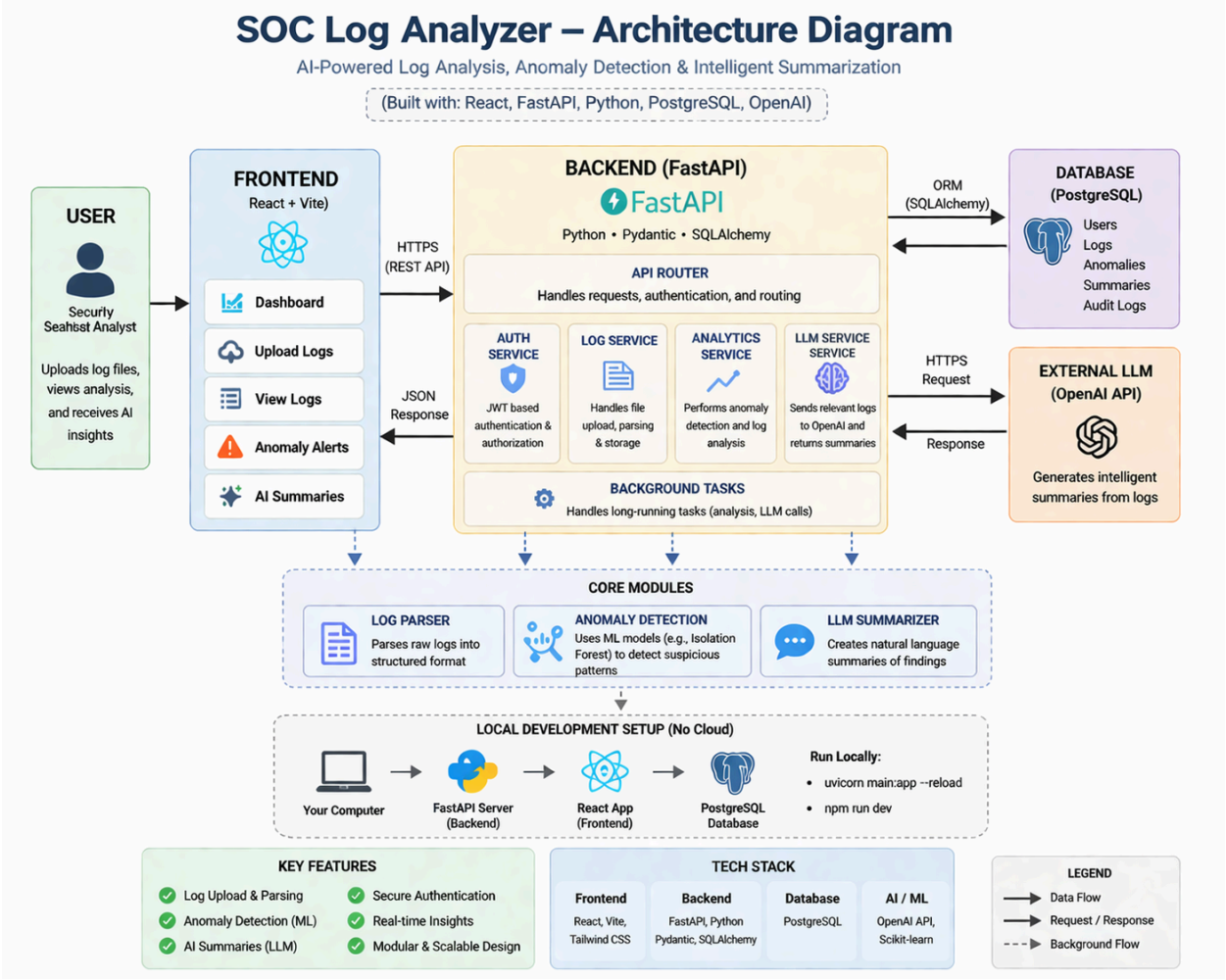
4. System Architecture

The system is built using a full-stack architecture:

- **Frontend:** React (User Interface & Visualization)
- **Backend:** FastAPI (API handling and processing)
- **Database:** PostgreSQL (for storing logs and analysis results)
- **AI Model:** OpenAI GPT (LLM for analysis and summarization)
- **Data Processing:** Pandas
- 

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5. Role of PostgreSQL (Important Section)

PostgreSQL plays a critical role in the system:

- Stores uploaded log data in structured tables
- Maintains processed and analyzed results
- Enables efficient querying of logs
- Supports scalability for large datasets
- Helps in tracking historical log analysis

Example Data Stored:

- Raw logs
- Parsed logs
- Anomaly detection results
- AI-generated summaries

👉 This ensures **data persistence and efficient retrieval**, which is essential for real-world systems.

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6. System Workflow

6.1 User Authentication

- Users log into the system securely
- Authentication ensures controlled access

6.2 Log Upload

- Users upload log files (.csv / .log)
- Files are sent to backend via API

6.3 Log Parsing (Preprocessing)

- Logs are converted into structured format
- Extracted fields:
    - IP, timestamp, URL, action

- Data is cleaned for analysis

6.4 Storage in PostgreSQL

- Parsed logs are stored in PostgreSQL
- Ensures:
    - Data persistence
    - Easy querying
    - Efficient processing

6.5 AI-Based Log Analysis (Core Component)

- Structured logs are sent to the AI model
- The model performs:
    - Pattern recognition
    - Behavior analysis
    - Threat classification

AI identifies:

- Suspicious IP behavior
- Repeated failed requests
- Malicious domains
- Unusual activity patterns

6.6 Anomaly Detection

- Logs are classified into:
    - Normal
    - Suspicious
    - Critical

Detection is based on:

- AI insights
- Frequency thresholds
- Behavioral anomalies

6.7 AI-Based Summarization

- AI generates:
  - Summary of events
  - Key threats
  - Security insights

👉 Converts raw logs into human-readable output

6.8 Output Visualization

- Results displayed in frontend:
  - Tables
  - Charts
  - Event timelines

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7. Role of AI in the System

Artificial Intelligence is used for:

- Log analysis
- Threat detection
- Anomaly identification
- Automated summarization

The AI model provides **context-aware intelligence**, improving detection accuracy over traditional systems.

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8. Key Features

- Upload and analyze log files
- Store logs in PostgreSQL
- Detect anomalies automatically
- AI-generated summaries
- Interactive dashboard
- Secure authentication

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9. Advantages

- Scalable data storage using PostgreSQL
- Detects unknown threats
- Reduces false positives
- Faster analysis and response
- Efficient handling of large log volumes

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10. Real-World Applications

- Enterprise security monitoring
- SOC operations
- Threat detection systems
- Cybersecurity analytics platforms

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11. Conclusion

The SOC Log Analyzer integrates **AI and PostgreSQL** to create a powerful cybersecurity solution that transforms raw log data into actionable insights. By combining structured data storage with intelligent analysis, the system improves efficiency, scalability, and accuracy in threat detection. This approach significantly reduces manual workload and enables faster and more effective security responses.

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12. Future Enhancements

- Real-time log streaming integration
- Advanced ML-based anomaly detection
- SIEM tool integration
- Role-based access control
- Multi-agent AI workflows